Release Notes

30 October 2026

I. ROS support

Version v1.0.1 adds optional ROS 1 and ROS 2 nodes for image and IMU topics. Both nodes use the same OpenVIO estimator and the same input processing code. Monocular and stereo configurations are available for EuRoC MAV and TUM-VI.

TABLE I
ROS DISTRIBUTIONS

ROS	Distribution	Ubuntu
ROS 1	Noetic	20.04
ROS 2	Humble	22.04
ROS 2	Jazzy	24.04

II. Separate builds

Use `./build_ros1.sh` for ROS 1. Use `./build_ros2.sh humble` or `./build_ros2.sh jazzy` for ROS 2. The two scripts run their respective build tools directly. Each distribution has its own output under `build_ros/<distro>/`. Source `build_ros/<distro>/install/setup.bash` in each terminal before use.

- The standalone build remains `./build.sh`, with output in `build/`.
- The ROS build does not call the standalone script or write to its output directory.
- Noetic uses a local Eigen 3.4 installation. The system Eigen installation stays unchanged.
- The ROS build stops if a dependency install, configuration step or compiler command fails.

III. Run and display

Run one command for the installed ROS distribution:

```
roslaunch openvio_ros1 openvio.launch dataset:=euroc
ros2 launch openvio_ros2 openvio.launch.py dataset:=euroc
```

Start the node before playback. For ROS 1 bags, use `rosbag play -d 2 <bag>`. Convert ROS 1 bags before ROS 2 playback.

The launch files open RViz with the estimated path, landmarks, tracked image and the `world` to `imu` transform. Use `stereo:=true` for a stereo dataset or `config:=/path/to/sensor.yaml` for another sensor. The optional `output_path` argument writes the IMU trajectory in TUM format.

IV. Verification and scope

ROS 1 Noetic and ROS 2 Jazzy were built and checked with EuRoC MH_01_easy monocular data. The checks confirmed a growing path, changing feature images, nonempty landmarks, and coordinate transforms. RViz displayed the outputs. These checks verify the ROS interface and display. They do not provide a new accuracy benchmark.

ROS 2 Humble is not tested. Stereo and live sensor input were not rerun in this release check. The accuracy results in the v1.0.0 release notes remain the reference results.

V. Documentation

The documentation links now follow this order: Quick start, ROS 1 / ROS 2, Release notes. See `doc/ros.md` for build commands, topic names, launch arguments, RViz controls and fault checks.